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Supporting Pharmacist-GP Collaboration in Medication Review Using Argumentation

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Abstract. Medication review is a collaborative process between pharmacists and general practitioners (GPs) aimed at optimizing patient care by identifying and eliminating harmful medications. This paper proposes a collaborative platform to enhance pharmacist-GP interactions, assess drug-drug interactions, evaluate adverse effects, and manage dosages. The platform uses the issue mapping function of IBIS to structure dialogues and systematically evaluates proposed actions using the QuAD framework to support decision-making. An ontology based on medical knowledge ensures consistency, while visual enhancements such as varying edge width, color coding, and highlighting preferred actions enable swift, informed decisions. These tools improve collaboration and patient care outcomes.

Keywords. Argumentation, IBIS, Issue mapping, QuAD, Debate, Decision support, Medication review, Visual Analytics.

1. Introduction

Medication review (MR) is a structured interview with the patient, conducted by the pharmacist in collaboration with the general practitioner (GP) to optimize patient care [1]. The review aims to identify and eliminate redundant or potentially harmful medications. Recommendations from the pharmacist are summarized and forwarded to the GP, who retains ultimate clinical decision authority. Effective collaboration between the pharmacist and the GP is crucial for maximizing patient health outcomes.

This work is part of a larger project, ABiMed [4,5], aimed at designing a clinical decision support system for MR. In this context, we propose here a collaborative chat-based platform to enhance exchanges between the pharmacist and the GP, and to facilitate decision making. This platform allows for the formal and visual structuring of the drug-related problems identified during MR, the proposed positions, i.e., solutions such as lowering dosage, and the pros and cons arguments. Using argumentation, the platform can systematically rank the positions.

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2. Methods

We applied the IBIS (Issue Based Information System [2]) and QuAD (Quantitative Argumentation Debate [3]) argumentation frameworks to the MR domain. IBIS structures complex dialogues and debates, particularly in contexts requiring comprehensive analysis of multifaceted issues. It facilitates the dissection and organization of arguments within a structured graphical model, effectively mapping out dialogues to enhance clarity and focus. It integrates relations indicating both support and opposition among arguments. QuAD provides automated decision support by quantifying the strength of positions, by aggregating the strength of their attacking and supporting arguments, providing a systematic quantification of the order of preference among various positions.

For example, let us consider a patient taking spironolactone, furosemide, ketoprofen, and dantrolene, who has severe renal failure. During MR, a drug-drug interaction (DDI) between spironolactone and furosemide is identified. Two solutions are possible: (p1) replace spironolactone with a calcium channel blocker, or (p2) stop ketoprofen. During the debate, the following arguments emerge: (a1) the DDI supports both solutions, (a2) NSAIDs such as ketoprofen should be avoided in cases of severe renal failure, supporting solution p2, and (a3) calcium channel blockers are contraindicated with dantrolene, attacking solution p1.

Using ABiMed tools [5], which contain structured drug knowledge and patient data, we propose an automated system that initially identifies key problems based on drug knowledge. The system then explores various appropriate solutions to address the identified problems using medical knowledge. Next, it defines some of the supporting and attacking arguments for each solution, and users can propose additional arguments. Then, we map the identified problems, solutions, and arguments using the **issue mapping** function of IBIS, a visual notation to map out the discourse:

- **Issues:** problems needing answers (e.g., the aforementioned DDI).
- **Positions:** Possible answers or solutions to the issues (e.g., p1, p2).
- **Arguments:** Statements supporting (pros) or attacking (cons) positions or other arguments (e.g., a1, a2, a3).

QuAD considers an initial weight for each argument and relation based on medical knowledge. It then recursively calculates the overall score of each position and argument through an aggregation method. The positions are ranked according to their scores.

To ensure consistency in the decision-making process, an ontology based on established medical knowledge was created. This ontology, in the context of MR, is a structured set of terms defining issues, positions, arguments, and their relations, such as supports and attacks. The ontology defines clear relations to maintain logical flow in discussions, specifying which arguments can logically support or attack which positions or other arguments.

Additionally, in the user interface, the ontology restricts the argument classes presented to the users to the consistent classes (Figure 1).

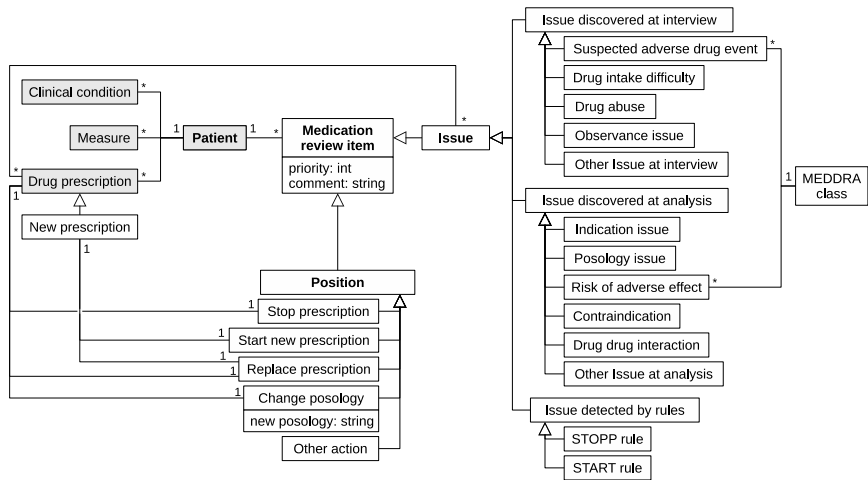


Figure 1. Ontology diagram illustrating the relations between various classes involved in the MR process. It includes entities such as patients, clinical conditions, drug prescriptions, and medication review items. Issues are categorized based on their discovery context (interview, analysis, or rules), and various positions are proposed to address these issues.

We visualize the issue mapping as a graph, using the following visual variables:

- **Edge width:** The width of each edge is proportional to the weight of the arguments and positions, helping visualize the reasoning.
- **Edge color:** Green indicates supporting relations, while red indicates attacking relations. Symbols (+ for support and - for attack) are added for colorblind users.
- **Node icon and color:** Preferred positions based on QuAD are highlighted with a distinct color and icon, allowing clinicians to immediately recognize the best position.

We implemented the proposed system as a web application, using Python with the Owlready [6] and Fullpy packages.

3. Results

We applied our system to the previous example, with the results illustrated in Figure 2. In this graph, the issue node is positioned at the top, and the position nodes are arranged directly below it. The optimal position is highlighted in green for easy identification, based on the aggregated score provided by QuAD, which is printed on each weighted node. Argument nodes are placed at the bottom, offering support or opposition to the positions.

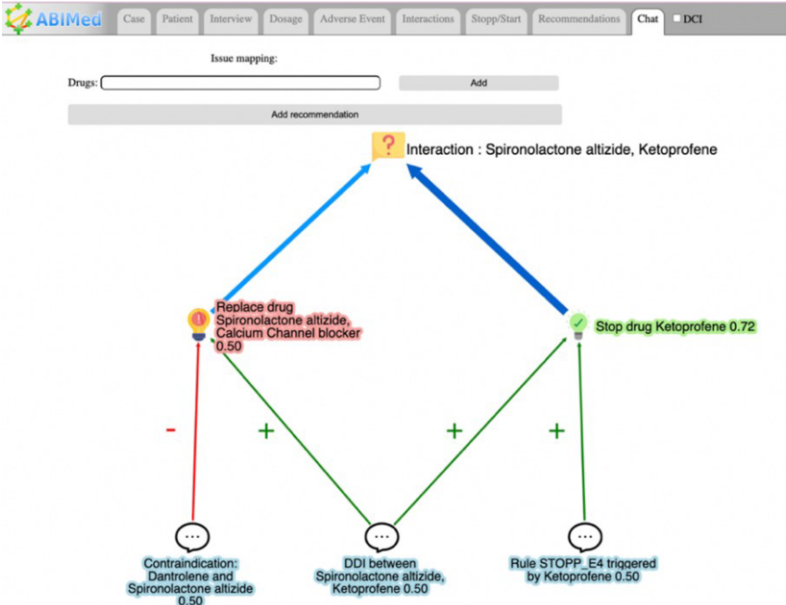


Figure 2. The corresponding issue mapping of the aforementioned example with one issue and two positions.

Users can interact with the graph by clicking on any node to display a popup dialog. If an issue node is clicked, the dialog prompts the user to add a position. Clicking on a position or argument node allows the addition of arguments that either attack or support the selected node. This interactive dialog leverages the ontology to filter the proposed positions based on the selected node, ensuring consistency throughout the graph.

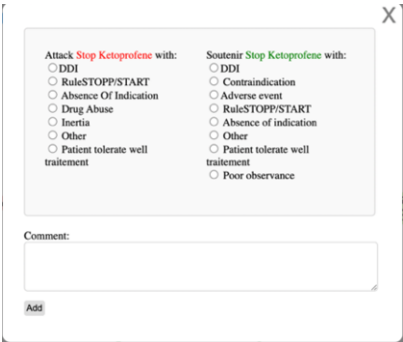


Figure 3. Screenshot of the pop-up dialog for adding new arguments and positions.

4. Discussion

We proposed a system to enhance collaboration between healthcare professionals in MR process, using IBIS and QuAD argumentation frameworks. The main limitation is the lack of a proper evaluation, necessary to validate its effectiveness and usability in clinical settings. While the system may require significant time from clinicians, it is particularly

useful for complex problems without clear solutions, especially in MR involving multiple actors where the pharmacist is not directly in front of the patient.

Additionally, QuAD's mathematical approach may not always capture the nuances of medical reasoning, even with an expert ontology and appropriate weights to bias the reasoning. Nonetheless, clinicians can use it as a tool for generating suggestions rather than definitive answers. Several decision support systems for MR have been developed [7], but none have incorporated argumentation. Integrating IBIS and QuAD into the MR decision-making process offers a novel approach for structuring and evaluating complex medical decisions.

5. Conclusions

In conclusion, our proposed system presents a promising tool for aiding medication review by structuring problems, evaluating solutions and arguments, and ensuring consistency through an ontology. Future work should focus on evaluating the system's effectiveness in clinical settings and exploring ways to integrate it seamlessly into clinicians' workflows to maximize its utility and impact. Additionally, further refinement of the ontology could help better align the system with medical reasoning, making it a more effective tool for clinicians.

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